

microeconomics_2_L5

Subject: #microeconomics

Topic: Price Discrimination — Unit Demand Setting

Definition. A consumer has **unit demand** for an indivisible good if their demand function is given by:

$$D(p) = \begin{cases} 1 & 0 \leq p \leq \omega \\ 0 & p > \omega \end{cases}$$

where $\omega \geq 0$ is the consumer's **willingness to pay (WTP)**.

Definition. In a market with N consumers, their WTPs can be ordered from highest to lowest:

$$\omega_1 > \omega_2 > \dots > \omega_N$$

Definition. The **socially optimal production level** q^* is the quantity that maximizes total welfare:

$$q^* = \arg \max_{k \in \{1, \dots, N\}} \left[\sum_{i=1}^k \omega_i - C(k) \right]$$

where $C(k)$ is the cost of producing k units.

Monopoly Pricing with Unit Demand

Definition. Under **linear (non-discriminatory) pricing**, a monopolist must charge a single price p to all consumers.

- To sell exactly k units, the price must satisfy $\omega_{k+1} \leq p \leq \omega_k$.
- The revenue-maximizing price for selling k units is $p = \omega_k$, yielding revenue $k \cdot \omega_k$.

The monopolist's profit-maximization problem is:

$$q_m = \arg \max_{k \in \{1, \dots, N\}} [k \cdot \omega_k - C(k)]$$

and the optimal price is $p_m = \omega_{q_m}$.

Note

The monopolist typically **under-produces** relative to the social optimum ($q_m < q^*$).

Definition. Under **first-degree price discrimination**, the monopolist can charge consumer-specific prices.

- To sell k units, the optimal prices are:

$$p_i = \begin{cases} \omega_i & \text{for } i \leq k \\ (\text{not served}) & \text{for } i > k \end{cases}$$

- Total revenue is $\sum_{i=1}^k \omega_i$.

- The monopolist **extracts all consumer surplus**.

The monopolist's problem becomes:

$$q_m = \arg \max_{k \in \{1, \dots, N\}} \left[\sum_{i=1}^k \omega_i - C(k) \right]$$

which is identical to the social planner's problem, resulting in $q_m = q^*$.

Note

$\sum_{i=1}^k \omega_i > k \cdot \omega_k$ for any $k \geq 2$, showing the revenue advantage of discrimination.

Class Example 1. Four consumers have WTPs:

$$\omega_1 = 10, \quad \omega_2 = 7, \quad \omega_3 = 6, \quad \omega_4 = 4$$

The monopolist's cost function is $C(q) = q^2$.

Questions:

1. Determine the socially optimal quantity q^* .
2. Find the monopolist's optimal price under linear pricing.
3. Find the optimal prices under first-degree price discrimination.

Second-Degree and Third-Degree PD with Unit Demand

Definition. In a **unit demand setting**, quantity-specific pricing provides **no discriminatory power** because consumers only buy one unit.

- **Second-degree price discrimination** (offering a single, non-linear tariff to all) is equivalent to linear pricing.
- **Third-degree price discrimination** (charging group-specific linear prices) allows the monopolist to set $p_i = \omega_i$ for each group i , which is equivalent to first-degree discrimination and extracts all consumer surplus.

Note

This result relies on the good being homogeneous. If the monopolist can differentiate products (e.g., by color or features), then **variety-specific pricing** can be used to discriminate effectively even within the same tariff structure.

Summary — Unit Demand Setting:

- **Consumer-specific prices** provide full discrimination power (1st-degree $\equiv$ 3rd-degree PD).
- **Quantity-specific prices** provide no extra power (2nd-degree PD $\equiv$ linear pricing).
- The critical assumption is a **homogeneous good**.

Sources:

Additional materials:

Review questions: